crossbar_simulation_log

May 9, 2026

1 Crossbar Fixed-Input Analysis

Measures column currents and floating voltages for a fixed input vector `v = [10, 20, 30, 40]`.

The current is measured at the bottom of each column with an ammeter with a resistance, R=.001 Ohms.

[40]:
```
import numpy as np

W        = np.array([[1,-1,1,-1],[1,1,-1,-1],[-1,1,1,-1],[-1,-1,-1,1]])
R        = 1.0
R_sense = 0.001

def measure_all_columns(v):
    """Ground each column in turn, return measured currents and floating␣
 ↪voltages."""
    v = np.array(v, dtype=float)
    results = {}
    Ip=np.zeros(4)
    for c in range(4):
        I = (W[:, c] * v).sum() / R / (1 + R_sense * np.abs(W[:, c]).sum() / R)
        Ip[c]=round(I,2)
        floats = {}
        for cf in range(4):
            if cf == c: continue
            SW   = W[:, cf].sum()
            SWV = (W[:, cf] * v).sum()
            if   abs(SW)  > 1e-10:  floats[cf] = f"{SWV/SW:+.2f}V"
            elif abs(SWV) < 1e-10:  floats[cf] = "0V"
            else:                   floats[cf] = "RAILS"
        results[c] = {'I': I, 'floats': floats}
    return results,Ip
```

[41]:
```
v = np.array([10, 20, 30, 40], dtype=float)
i_ideal = W.T @ v
res,Ip = measure_all_columns(v)

print(f"Input: v = {v.astype(int).tolist()}")
```

```python
print(f"Ideal output (W.T @ v): {i_ideal.astype(int).tolist()}")
print("simulated output", Ip)
print()
print(f"  {'Col':>4}  {'I_sense':>9}  {'I_ideal':>9}  {'Error':>9}  Floating␣
↪voltages")
print(f"  {'-'*70}")
for c in range(4):
    I      = res[c]['I']
    floats = res[c]['floats']
    fstr   = "  ".join(f"V_col{cf}={fv}" for cf, fv in floats.items())
    print(f"  {c:>4}  {I:>9.4f}  {i_ideal[c]:>9.4f}  {I-i_ideal[c]:>9.6f} ␣
↪{fstr}")
```

```
Input: v = [10, 20, 30, 40]
Ideal output (W.T @ v): [-40, 0, -20, -20]
simulated output [-39.84   0.   -19.92 -19.92]

   Col    I_sense    I_ideal      Error  Floating voltages
  ----------------------------------------------------------------------
     0   -39.8406   -40.0000   0.159363  V_col1=0V  V_col2=RAILS  V_col3=+10.00V
     1     0.0000     0.0000   0.000000  V_col0=RAILS  V_col2=RAILS
V_col3=+10.00V
     2   -19.9203   -20.0000   0.079681  V_col0=RAILS  V_col1=0V  V_col3=+10.00V
     3   -19.9203   -20.0000   0.079681  V_col0=RAILS  V_col1=0V  V_col2=RAILS
```